# SURJO DEY

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## EDUCATION

**Rajiv Gandhi Institute of Petroleum Technology**, Raebareli, India **2022 – 2026**
B.Tech in Computer Science and Engineering *CGPA: 3.22/4*
Recipient of the Government of India's Study in India (SII) Scholarship Program *WES U.S. Equivalency*

## RESEARCH EXPERIENCE

**Deep Intelligence Lab, Prince Mohammad Bin Fahd University**, Saudi Arabia **Oct'26 – Present**
*Junior Research Associate — Supervisor: Dr. Muhammad Attique Khan* *Remote*
Working on Vision-Language Models (VLMs), Specially on JEPA (Joint-Embedding Predictive Architecture) and Hyperspectral image dataset.

**Tezpur University**, Assam, India **Mar'26 – Sep'26**
*Research Fellow — Supervisor: Dr. Swalpa Kumar Roy* *Remote*
Worked on Vision-Language Models (VLMs), build better ways to align image and text representations for stronger visual grounding and understanding across multimodal tasks.

**Jashore University of Science and Technology**, Jashore, Bangladesh **Jun'25 – Sep'25**
*Research Intern — Supervisor: Dr. A F M Shahab Uddin* *On-site*
Developed segmentation-guided diffusion models for medical image augmentation using the Duke-Breast-Cancer-MRI dataset to improve robustness and variability in diagnostic model training data.

## PUBLICATIONS

A. Das, **S. Dey**, K. Biswas, S. K. Roy, M. Abdar, V. K. Verma. "**SALD: Self-Referenced Advantage Learning for Diffusion Models**". *Submitted:* ***ICLR***, 2027 [Preprint]
Proposed a training framework that improves image generation by comparing predictions at different noise levels using a shared model, without requiring an external teacher network.

**S. Dey**, P. Saikia. "**Explainability in Generative Medical Diffusion Models: A Faithfulness-Based Analysis on MRI Synthesis**". *3rd World Conference on Smart Computing Systems (WCSC)*, 2026 [Paper]
Proposes a faithfulness-based explainability framework using prototype methods to make generative diffusion models transparent for breast MRI synthesis.

**S. Dey**, A. Sharma, H. Raj, S. Biswas. "**Automated Detection and Analysis of Minor Deformations in Flat Walls Due to Railway Vibrations Using LiDAR and Machine Learning**". *International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2024 [Paper]
Proposes an automated LiDAR and machine learning pipeline to detect and quantify minor structural wall deformations induced by railway vibrations in urban infrastructure.

**S. Dey**, A. Sharma, H. Raj, S. Biswas. "**Comprehensive Analysis of Structural Defects in Various Structures Using TLS Data and Machine Learning**". *IEEE International Conference on Future Machine Learning and Data Science (FMLDS)*, 2024 [Paper]
Develops an automated ML-based system using TLS point cloud data to detect and analyze structural deformations across various building geometries.

*Some other Papers are under review on Core Ranked Conferences*

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, MATLAB, C |
| **ML / DL** | PyTorch, Diffusers, Transformers, Scikit-learn, Pandas, NumPy, Matplotlib |
| **Research** | Diffusion Models, Vision-Language Models, Fine-tuning, Curriculum Learning |
| **Tools** | LaTeX, Git, Jupyter, VS Code, Google Earth Engine |
| **OS** | Linux (Ubuntu), Windows |

## REFERENCES

**Dr. Pallabi Saikia**
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RGIPT, Raebareli, India
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**Dr. Swalpa Kumar Roy**
Associate Professor
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**Dr. Vinay Kumar Verma**
Senior Applied Scientist
Amazon, USA
Formerly Duke University, USA
vkvermaa@amazon.com